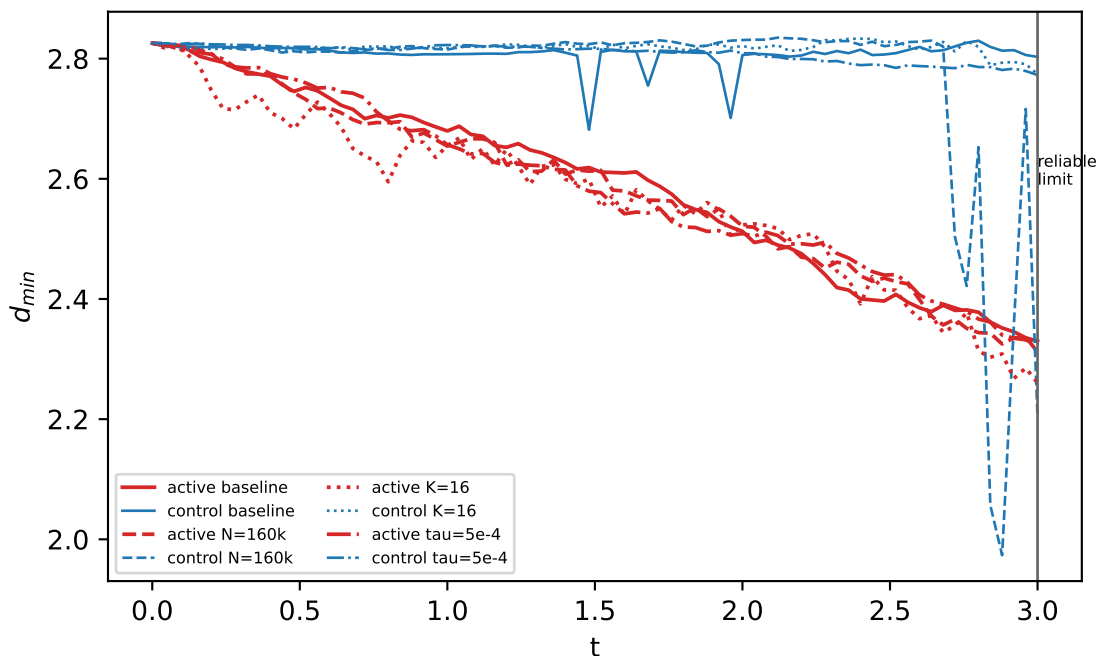
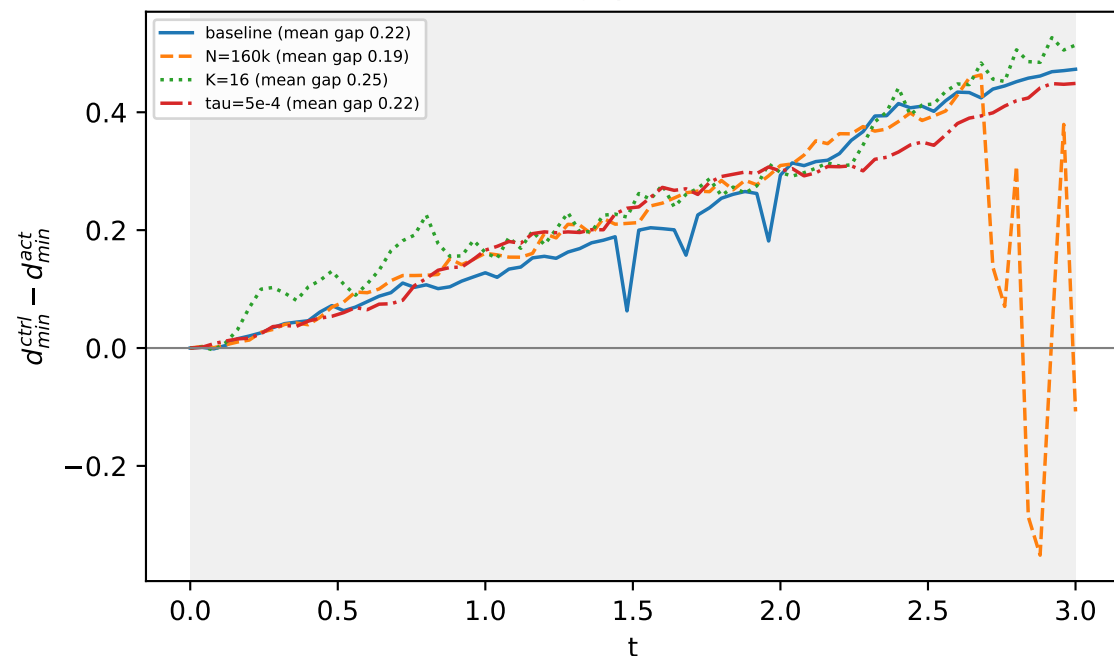


Tetra resolution / centroid-reliability validation (M=240, a=1.0; one-factor N/K/tau refinement)

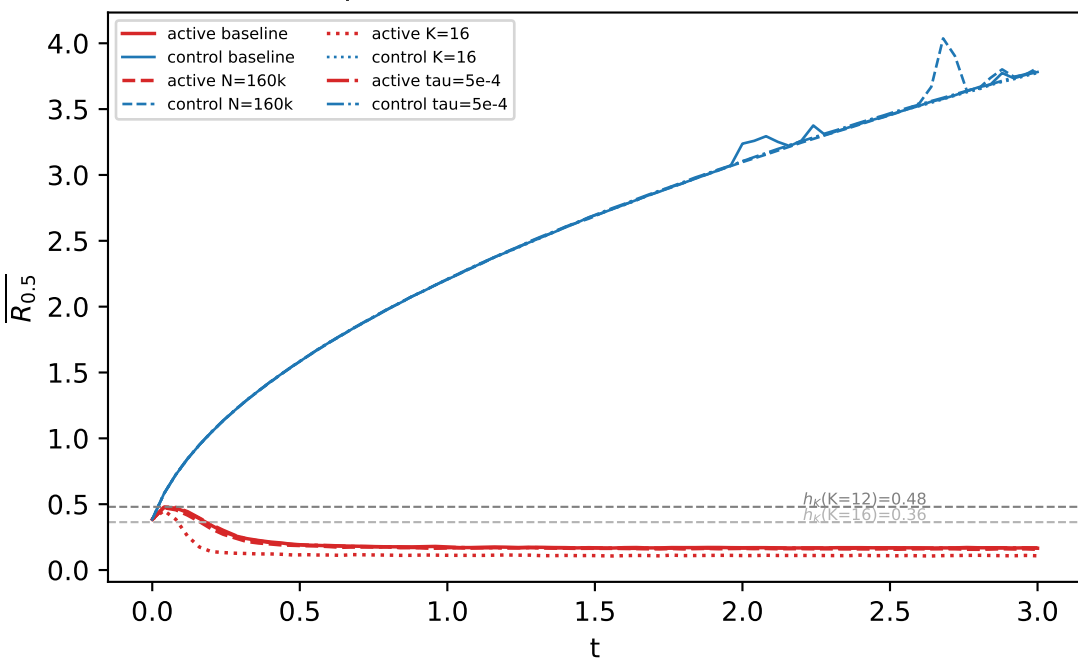
(a) $d_{min}(t)$ active (red) vs control (blue)



(b) active-control d_{min} gap (shaded=reliable interval)



(c) mean per-cluster $R_{0.5}(t)$ (field scale h_K marked)



(d) centroid reliability $A_{min}(t) = \min_m \min_j A_{m,j}$

